Topic1-2

July 5, 2026 2:14 PM

Inference in SLR

What do you know about the true population line?

Photo by **Nathan Dumlao** on **Unsplash**

Announcements

- worksheet_01 and tutorial_01 are due on Saturday, July 5, at 11:59pm.

Participate and Collaborate!

- Some classes will have iClickers polls. Partial marks for correctness, rest for participation, with weight depending on the poll.
- Talk to each other (in class, on Piazza) as you work through the worksheets and tutorials

In [5]:

```
# packages
library(tidyverse)
library(repr)
library(infer)
library(cowplot)
library(broom)

#data
dat <- read.csv("data/Assessment_2015.csv")
dat <- dat %>% filter(ASSESSCLAS=="Residential")  %>%
        mutate(assess_val = ASSESSMENT / 1000)
```

$y = f(x) + \varepsilon$

$\beta_0 + \beta_1 x$

Last class: the regression framework

- Linear Regression Models provide a unifying framework to estimate and test the true relation between different type of variables and a **continuous response**
- Linear Regression Models can also be used to predict the value of a **continuous response** (although it may not be the best predictive model)
- You assume that the conditional expectation of the response is linearly related to the input variable and the line is the **linear regression**

Last class: estimation

- The population regression coefficients (β_0 and β_1) are *unknown* and *non-random*
- You use a random sample to estimate them
- **Least Squares** is one method to estimate the coefficients of a regression: it minimizes the sum of squares of the residuals
- In R, LS regression estimates can be computed using the `lm` function
- Using data from observational studies, we can not study any cause or effect between variables, only associations.

$\sum_{i=1}^{n}(y_i-(\beta_0+\beta_1 x_i))^2$

residual

linear model

β_1

Today: Inference

$\beta_1 = ?$

Using data, you can obtain point estimates of the regression coefficients. But,

- how can you infer information about the *population regression parameters* using those *estimates*?
- your estimates depend on your sample. How much sample-to-sample variation do you expect?

Vancouver: $\hat{\beta}_1 = 2.3$

Toronto: $\hat{\beta}_1 = 2.1$

$\hat{\beta}_1 = 2.7$

...

Estimators of the regression coefficients

The LS regression estimators are functions of the random sample. Thus, they are also random variables.

- We call them $\hat{\beta}_0$ and $\hat{\beta}_1$ to differentiate them from the population coefficients.
- As any other random variable, each estimator has a distribution. Since they are statistics, these distributions are called *sampling distributions*.
- As any other random variable, each estimator has a standard deviation. Since they are statistics, their SDs are called *standard errors* (SE).
- Both the SE and the sampling distribution are needed to make *inference about the population coefficients*.

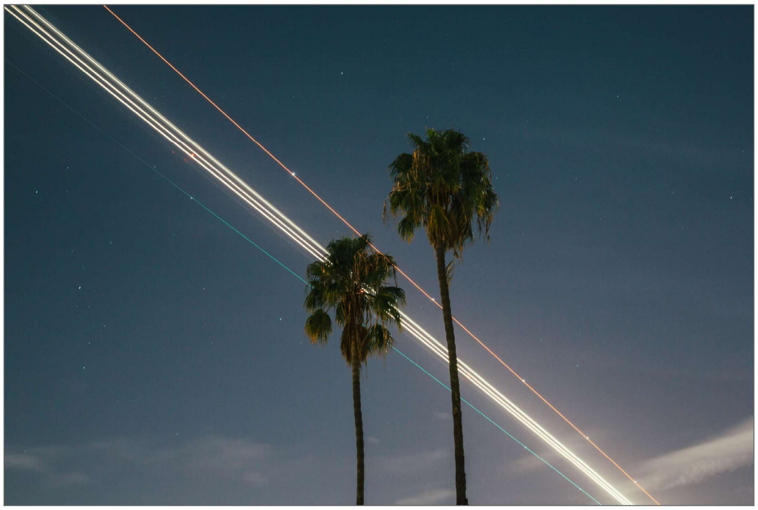
Q: What's the difference between SD and SE?

$SD(\bar{y}) = \frac{\sigma}{\sqrt{n}}$ σ is unknown

$SE(\bar{y}) = \frac{\hat{\sigma}}{\sqrt{n}}$

Measuring variation

DIFFERENT SAMPLES RESULT IN DIFFERENT ESTIMATES!! BUT HOW MUCH VARIATION DO YOU EXPECT?



Let's look at an example!
We will use a dataset on house prices, containing 27699 observations as our **population**.

In [9]:

```
nrow(dat)
```

27699

We will randomly sample 1000 houses, and based on which fit a linear regression model for assessment value with building size as input variable. Here's estimate we obtain with this sample of 1000 houses `dat_s` from all properties in Strathcona County in Alberta `dat`:

In [10]:

```
set.seed(561)

dat_s <- sample_n(dat, 1000, replace = FALSE)

lm_value <- lm(assess_val~BLDG_METRE,data=dat_s)%>%
                tidy()%>%mutate_if(is.numeric, round, 3)
many_SLR <- lm_value %>% select(term,estimate)
```

$$y = \underline{\underline{\beta_0}} + \underline{\underline{\beta_1}} x + \varepsilon$$

In [11]:

```
many_SLR
```

A tibble: 2 × 2

term	estimate
<chr>	<dbl>
(Intercept)	90.769
BLDG_METRE	2.618

one sample

β_0

β_1

Take *another sample* from the full dataset and estimate the regression line again!

> ***Important***: *In practice, you will rarely take multiple samples!*

> ***NOTE***: *This is NOT bootstrapping!! Why??*

In [32]:

```
many_SLR
```

A tibble: 2 × 3

term	estimate	estimate2
<chr>	<dbl>	<dbl>
(Intercept)	90.769	28.833737
BLDG_METRE	2.618	3.017228

$y = \beta_0 + \beta_1 x + \varepsilon$

In [34]:

```
many_SLR
```

A tibble: 2 × 4

term	estimate	estimate2	estimate3
<chr>	<dbl>	<dbl>	<dbl>
(Intercept)	90.769	28.833737	117.497738
BLDG_METRE	2.618	3.017228	2.447751

$\hat{\beta}_0$

$\hat{\beta}_1$

3 samples from the same population

Q: Which one is the best estimate?

A: We don't know

$$SE(\hat{\beta}_1) = \sqrt{\frac{1}{3-1}\sum_{i=1}^{3}\left(\hat{\beta}_{1i} - \bar{\hat{\beta}}_{1i}\right)^2}$$

AND SO ON .... AS YOU TAKE NEW SAMPLES YOU GET DIFFERENT *ESTIMATES* OF THE REGRESSION PARAMETERS

The standard error!!

The variation of these estimates from sample to sample is measured by their standard deviation, which has a special name: *the standard error* (SE)

> *But in practice, how can we compute the standard error if we have only 1 sample??*

We have different ways of answering this question:

1. use a theoretical result! This is what `lm` does!!
2. use bootstrapping!! As you did in STAT 201 for other quantities!!

Inference

In [35]:

```
lm_value
```

A tibble: 2 × 5

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	90.769	9.793	9.268	0
BLDG_METRE	2.618	0.059	44.514	0

β_1 $SE(\beta_1)$

95% C.I. for β_1 :

$2.618 \pm 1.96 \times 0.059$

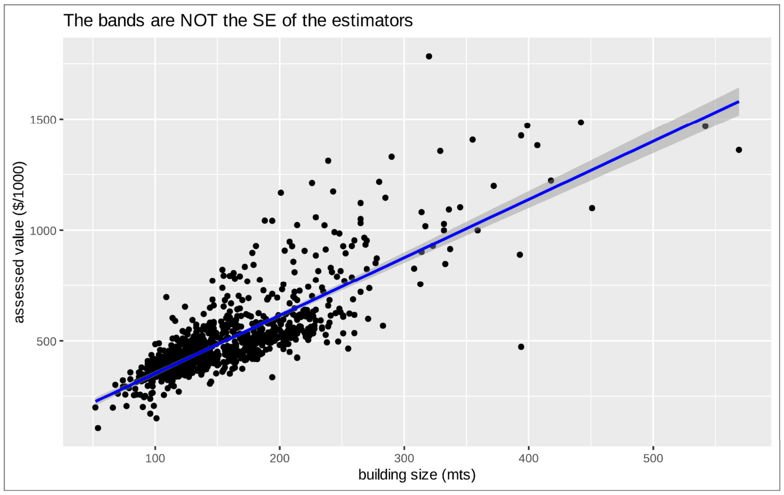
$(2.618 - 1.96 \times 0.059,\ 2.618 + 1.96 \times 0.059)$

Caution: these SE measure the sample-to-sample variation of each estimate, not the predicted value from the line. Thus, you can't visualize them in a usual scatter plot.
Choosing `se=TRUE` in `geom_smooth` you get bands for the predicted values

In [6]:

```
plot_value
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Hypothesis Tests

$y = \beta_0 + \beta_1 x + \varepsilon$

β_1 key parameter.

Is the input variable linearly associated with the response?

	Intercept	Slope
null hypothesis H_0 :	$\beta_0 = 0$	$\beta_1 = 0$
alternative hypothesis H_1 :	$\beta_0 \neq 0$	$\beta_1 \neq 0$

NOTE THAT WE HAVE SEPARATE TESTS FOR EACH PARAMETER IN THE REGRESSION

Test $H_0 : \beta_1 = 0$ (no linear relationship between x and y)

$H_1 : \beta_1 \neq 0$ (there is a linear relationship between x and y)

The statistic

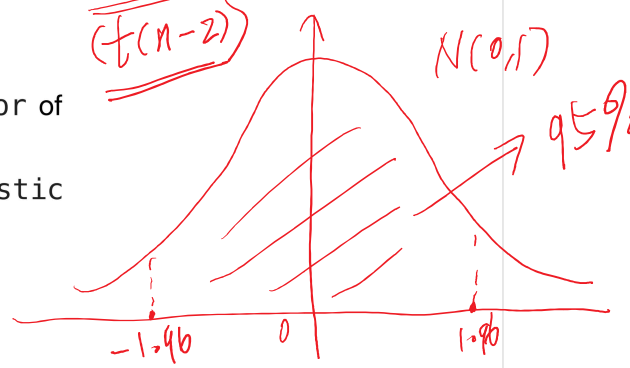
You can use the estimated coefficients and check how far the estimate is from 0.

> *how far is far??*

$t(n) \to N(0,1)$ if $n \to \infty$

The standard error can be used to construct a statistic. For example, the statistic of the slope is

$$T = \frac{\hat{\beta}_1 - 0}{\text{SE}(\hat{\beta}_1)} = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} \sim \frac{N(0,1)}{(t(n-2))} \text{ under } H_0$$



- an estimate of the SE is given in the column `std.error` of `tidy` table
- the value of this statistic is given in the column `statistic` of `tidy` table

$T=3$

The p-value

In the `tidy` table, you can also find p-values in the column `p.value`.

> *how are these p-values computed?? and for which hypotheses??*

You need the sampling distributions (distribution of the estimators $\hat{\beta}_0$ and $\hat{\beta}_1$) to compute p-values!!

In [44]:

```
lm_value
```

A tibble: 2 × 5

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
$\hat{\beta}_0$ (Intercept)	90.769	9.793	9.268	0
$\hat{\beta}_1$ BLDG_METRE	2.618	0.059	44.514	0

$$T = \frac{2.618}{0.059}$$

Caution: the alternative hypothesis in `lm` is $H_1 : \beta_j \neq 0$, for all j th coefficients.
The `p.value` is interpreted as the probability, under H_0, that $|T|$ is equal or larger than the value observed in your sample
For example, assuming the true slope is zero, the probability of observing a estimated slope as large or larger than 2.618 in *absolute value* is less than 0.001.

Decision rule

> *The smaller the p-value, the stronger the evidence against H_0.*

A small p-value (less than the significance level α) indicate that the data provide enough statistical evidence against the null hypothesis of no association (i.e., to reject H_0).
At a significance level of $\alpha = 0.01$ we reject H_0.

> *The true slope is different from 0 with strong evidence.*

Heads up: in the last years, the scientific community has identified the "crisis of p-values". You can read more about it in **this article** and in the **ASA statement**.

Confidence Intervals

While point estimates provide a single guess about the population coefficients, you may prefer to have a range of values

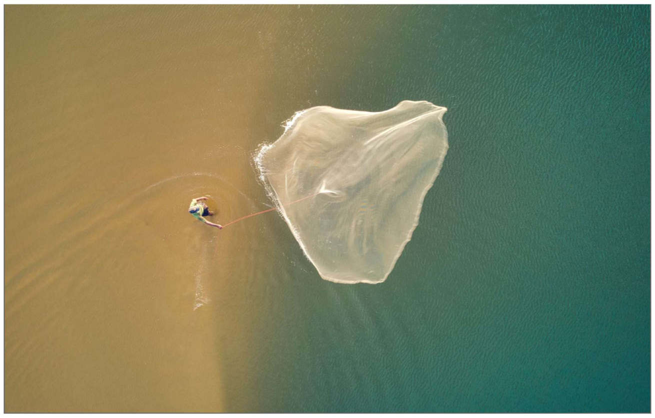


Photo by **lahiru iddamalgoda** on **Unsplash**. Analogy from **ModernDive**

Two ways for inference of β_1

equivalent (① 95% C.I. for β_1

② Test hypotheses: $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$ ($\alpha = 0.05$)

Q: What is the relationship between C.I. and hypothesis testing?

A: 95% C.I. of β_1 covers 0 $\Longleftrightarrow$ fail to reject H_0 at 5% level.

Classical CIs

estimate $\pm$ $Z^* \cdot SE$(estimate)

$$\left(\hat{b} - \mathrm{SE}(\hat{b}) \times t_{\alpha/2,n-k}, \hat{b} + \mathrm{SE}(\hat{b}) \times t_{\alpha/2,n-k}\right)$$

$t_{\alpha/2,n-k} \to z_{\frac{\alpha}{2}}$

where

- $\mathrm{SE}(\hat{b})$ is the (estimated) SE of the estimator
- n is the sample size and k the number of regression parameters
- The $t_{\alpha/2,n-k}$ is the **quantile of the t-distribution** with $n-k$ degrees of freedom!!

Heads up: this result is based in classical theoretical approximations (CLT) or distributional assumptions!!

Note: if the conditional distribution of the error term is assumed $N(0, \sigma^2)$ with σ known, then we use the **quantile of the Normal distribution** instead (i.e., $z_{\alpha/2}$ instead of $t_{\alpha/2,n-k}$)

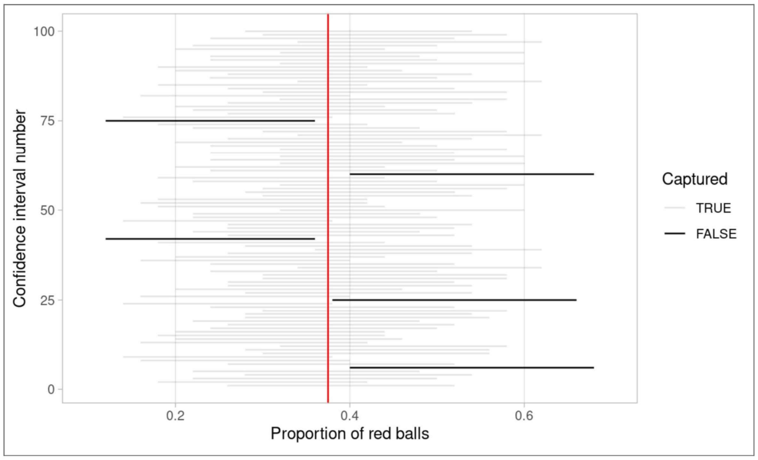
β_1

There are also many missunderstandings around the concept of CIs
A 95% CI computed from the data is **not** a range of values that contains the true regression parameter with 95% probability.

> *Once the interval has been computed based on the data, nothing is random! so it either covers or not the true value.*

Among many CIs computed from different samples, 95% of them contain the true regression parameter. Thus, we are 95% *confident* that the true coefficient is in the given range.

Recall from ModernDive:



In [51]:

```
lm_value_CIs
```

A tibble: 2 × 7

term	estimate	std.error	statistic	p.value	conf.low	conf.high
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	90.769	9.793	9.268	0	71.551	109.987
BLDG_METRE	2.618	0.059	44.514	0	2.502	2.733

The Sampling Distribution

We mentioned before that the estimators of the regression coefficient, $\hat{\beta}_0$ and $\hat{\beta}_1$, are *random variables*.
Then they also have a *distribution*, called the *sampling distribution*, which we can use to compute p-values!!

> *how do we know the sampling distribution of the estimators of the regression coefficients??*

Need assumptions, such as normal distribution.

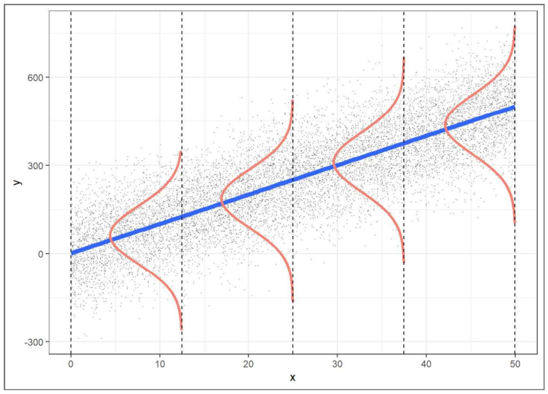
We have different ways of answering this question:

1. use a theoretical result! This is what `lm` does!!
2. use bootstrapping!!

Need no assumptions, no formulas.

Confirm the theoretical result.

THEORETICAL RESULTS



Images from **[Beyond Multiple Linear Regression (BMLR)]**

In regression, it is usually assumed that the conditional distribution of the error terms is Normal!!

Classical Theory 1: if we assume that the (conditional) distribution of the error terms is Normal, under H_0, the statistic T follows a t-distribution with $n - k$ degrees of freedom, where n is the sample size and k the number of regression parameters

Important: **CLT to the rescue**: if the assumption is not true but the conditional distribution of the error terms is *nice enough* (under certain conditions), and if the *sample size is large*, then **Classical Theory 2**: under H_0, the Central Limit Theory can be used to prove that the statistic T follows *approximately* a t-distribution with $n-k$ degrees of freedom

> *This theoretical result is used by* `lm` *to compute p-values and confidence intervals!!*

`lm` approximates the sampling distribution with a t-distribution with $n-k$ degrees of freedom

Stat 306

Math 220

Stat 305.

Stat 460/461

Math 320 real analysis

420

Bootstrapping

Photo by **Maxim Hopman** on **Unsplash**

Stat 200
Stat 201
Stat 306
Stat 301
Coding in R

> ***Historical note***: *The term bootstrapping originates from the phrase "to pull oneself up by one's bootstraps", which refers to completing a seemingly impossible task with no external help.*

The collected sample acts as the "bootstraps" that you can use to "pull yourselves up" to approximate of sampling distribution.

1
- You use the original sample as an *estimate* of the unknown population

2
- You sample from your original sample **with replacement** to generate a new sample of same size!

3
- To obtain different samples of equal size, you need to sample **with replacement**

Again, note that we are sampling from the sample!! not from the population!!

Bootstrapping

Bootstrap sample ⟶ Sample ⟶ Population

$(x_1^*, y_1^*), \dots (x_n^*, y_n^*): \hat{b}^{*1}$

.
.
.

$(x_1^*, y_1^*), \dots (x_n^*, y_n^*): \hat{b}^{*B}$

$(x_1, y_1), \dots (x_n, y_n): \hat{b}$ $D(b)$

- estimate the sampling distribution of $\hat{b}$
- estimate the mean and SE $\hat{b}$

- estimate the true *b*

The population is to the sample as the sample is to the bootstrap sample

Q: What is the SE of $\bar{y}$?

A: $SE(\bar{y}) = \frac{s}{\sqrt{n}}$

$s = \hat{\sigma}$

Q: What is the SE of ~~correlation r~~ median?

A: use bootstrap.

Using *bootstraping*, we generate a *long* list of estimates to compute *empirically* the sampling distribution! → $B = 100,\ 1000$

- sample with replacement to obtain B samples of size n
- for each sample, compute the estimated regression coefficients
- use the B regression estimates to calculate the sampling distribution → SE

In [11]:

```
head(lm_boot)
```

A data.frame: 6 × 2

	boot_intercept	boot_slope
	<dbl>	<dbl>
1	77.60172	2.734390
2	91.29174	2.604643
3	70.83473	2.798641
4	77.41625	2.726911
5	108.46348	2.454075
6	119.70836	2.420410

Boot1

Boot2

$\hat\beta_0$

$\hat\beta_1 = \frac{2.734390 + 2.604643 + \cdots}{6}$

Bootstrap estimate of β_1

$B = 6$

Bootstrap SE of $\hat\beta_1$

$SE(\hat\beta_1) =$ Sample SD of $(2.734390, 2.604643, \ldots, 2.420410)$

95% C.I. for β_1 is $\hat\beta_1 \pm 1.96\, SE(\hat\beta_1)$.

$H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$.

Test stat $Z = \frac{\hat\beta_1}{SE(\hat\beta_1)} \sim N(0,1)$

$B \geq 100$

In [12]:

```
tail(lm_boot)
```

A data.frame: 6 × 2

	boot_intercept	boot_slope
	<dbl>	<dbl>
9995	67.00700	2.772294
9996	102.80346	2.514296
9997	87.62634	2.638956
9998	81.86622	2.671669
9999	77.52043	2.719633
10000	86.87740	2.617274

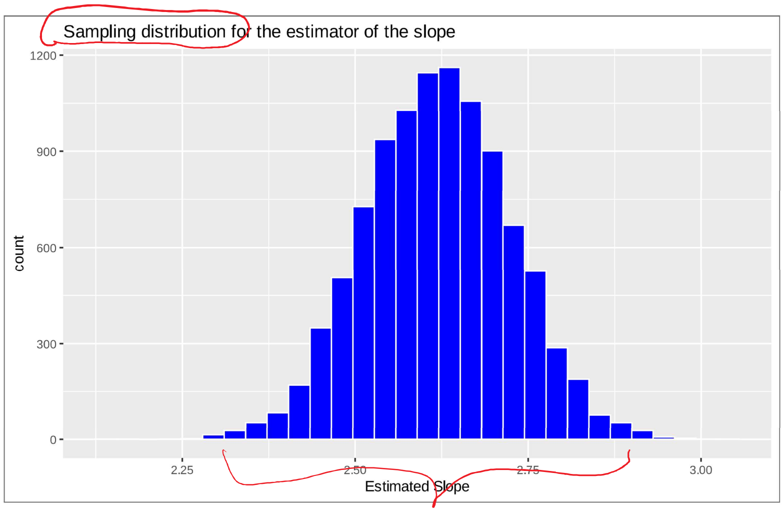
We generated 10000 estimates of the intercept and the slopes using only 1 sample!!!

We can visualize the distribution of these values!

In [14]:

```
slope_sampling_dist
```

```
`stat_bin()` using `bins = 30`. Pick better value with `b
inwidth`.
```



This is an approximation of the sampling distribution

① Frequentist ← likelihood methods

② Bayesian

iteratively algorithms

$B \geq 100$

EM algorithm

Does the number of replicates matter?

The larger the replication, the longer the list of estimates to better approximate of the sampling distribution! The approximated sampling distribution becomes smoother.

Does the sample size matter?

Our estimator depends on a random sample and thus on its size!

- The SE of the estimator decreases with the sample size. Thus, the sampling distribution becomes tighter
- From the CLT: the sampling distribution becomes more "bell-shaped as the sample size increases.

data

$f(y|\theta)$ ← likelihood

parameters

Bayes: $f(\theta|y) = \dfrac{f(y|\theta)f(\theta)}{f(y)}$

$\int f(y|\theta)f(\theta)\,d\theta$

MCMC

100

t-test

$\sqrt{n}(\bar{y}-\mu) \xrightarrow{d} N(0, \sigma^2)$ CLT

$\sqrt{n}(\bar{y}-\mu) \xrightarrow{d} N(0, \sigma^2)$ CLT
as $n \to \infty$

Bootstrap Confidence Intervals

Bootstrap methods work for any sample size, any distribution, no matter how complicated the estimate may be.

You can use the bootstrapping sampling distribution to compute CIs of regression parameters. There are various methods:

- **Standard error method**: use the list of bootstrap estimates to approximate (only) the SE of the estimator

$$\left(\hat{\beta}_1 - z_{\alpha/2}\mathrm{SE}^*, \hat{\beta}_1 - z_{1-\alpha/2}\mathrm{SE}^*\right)$$

- **Percentile method**: use the list of bootstrap estimates to the range using quantiles

$$\left(\hat{\beta}^*_{j,\alpha/2}, \hat{\beta}^*_{j,1-\alpha/2}\right)$$

95% Bootstrap C.I. (a, b)

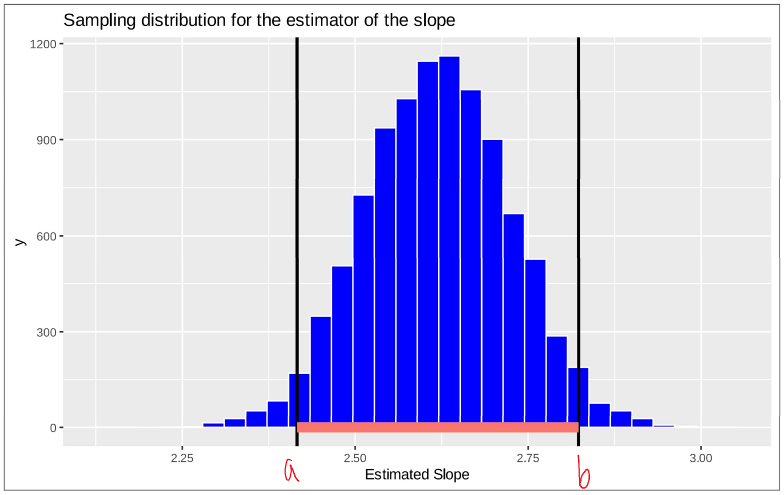
Bootstrap estimates $(B=100)$

a b
$* * * * \ldots * * * *$
$a = 2.5$ percentile $b = 97.5$ percentile

In [16]:

```
slope_sampling_dist
```

```
`stat_bin()` using `bins = 30`. Pick better value with `b
inwidth`.
```



$H_0: \beta_1 = 0$

$H_1: \beta_1 \neq 0$

reject H_0 at 5%